Lesson 09: How do we generate electricity?

Name: _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 0.00

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? -0.21 to 0.25

3. Talk with your partner. Why do you think that happened?

When you give it bonce the more eletcity it creates.

Papahua Skatepark

1. Keep the motor still. What does the meter read? 00.0

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

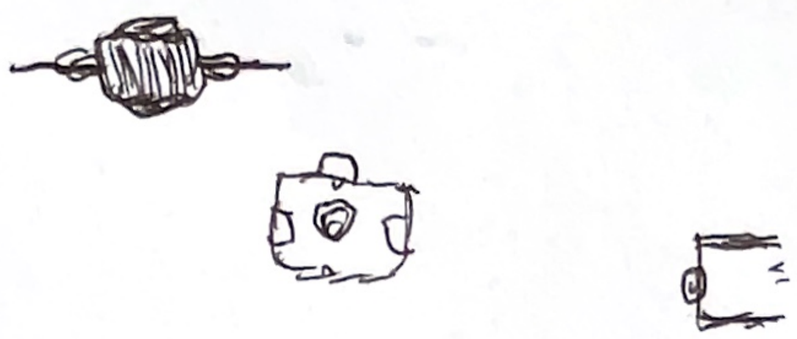
Slow: 3 to 0 Vaultes Faster: 180 to 0

3. Talk with your partner. Why do you think that happened?

When you spin it the Vaultage increase depending on how fast you spin it.

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

the wires coiled and magnets

2. What moved at Ngārunui Beach? magnets and weights

What turned at Papahua Skatepark? the motor

3. Talk with your partner. Why do you think the meter changed in both tests?

the voltage changed because of the magnet made electric.

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2. Gently push the weights down a little, then let go. What does the meter do as the weights move? -0.21 to 0.25

3. Talk with your partner. Why do you think that happened?

The more it bounces the more electricity it generates

Papahua Skatepark

1. Keep the motor still. What does the meter read? 00.0

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

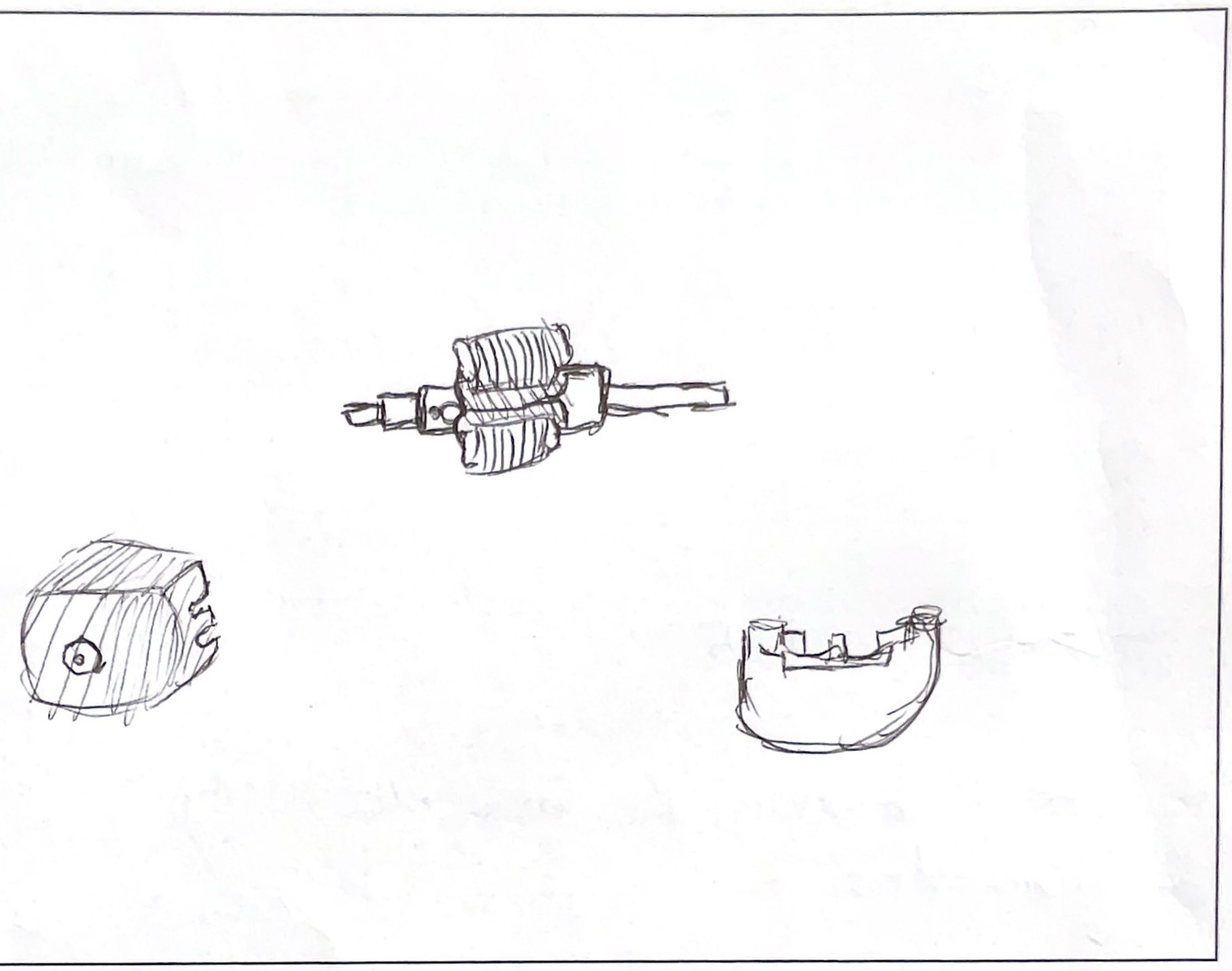
Slow: 3 to 0 volts Faster: 180 to 0

3. Talk with your partner. Why do you think that happened?

When you spin it the voltage increases depending on your speed

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



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2. Gently push the weights down a little, then let go. What does the meter do as the weights move? -0.21 to 0.25

3. Talk with your partner. Why do you think that happened?

The more it bounces the more electricity it generates

Papahua Skatepark

1. Keep the motor still. What does the meter read? ~~00.0~~ 000.0

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

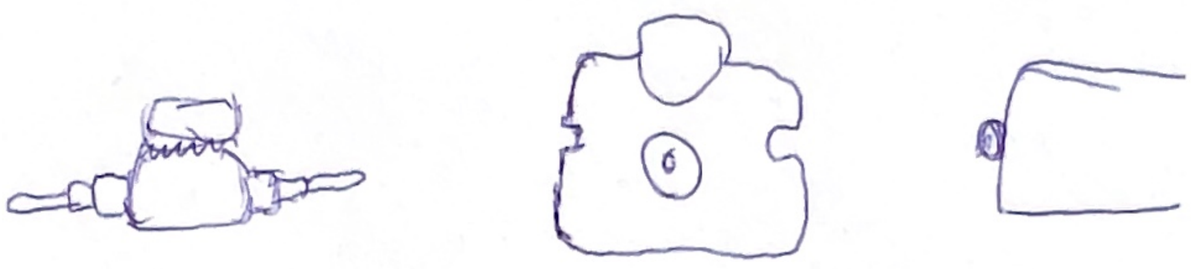
Slow: 3 to 0 volts Faster: ~~180~~ 180 to 0

3. Talk with your partner. Why do you think that happened?

~~When you~~ The faster you spin the higher the voltage

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

the wires coiled and magnets

2. What moved at Ngārunui Beach? the magnets and weights

   What turned at Papahua Skatepark? the motor

3. Talk with your partner. Why do you think the meter changed in both tests?

The voltage changed because the magnet made electricity.

Lesson 09: How do we generate electricity?

Name: _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 00.0

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? Changes

3. Talk with your partner. Why do you think that happened?

Magnets are ~~Changing~~ Voltage ~~Something~~ in the wires Creating

Papahua Skatepark

1. Keep the motor still. What does the meter read? 00.0

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

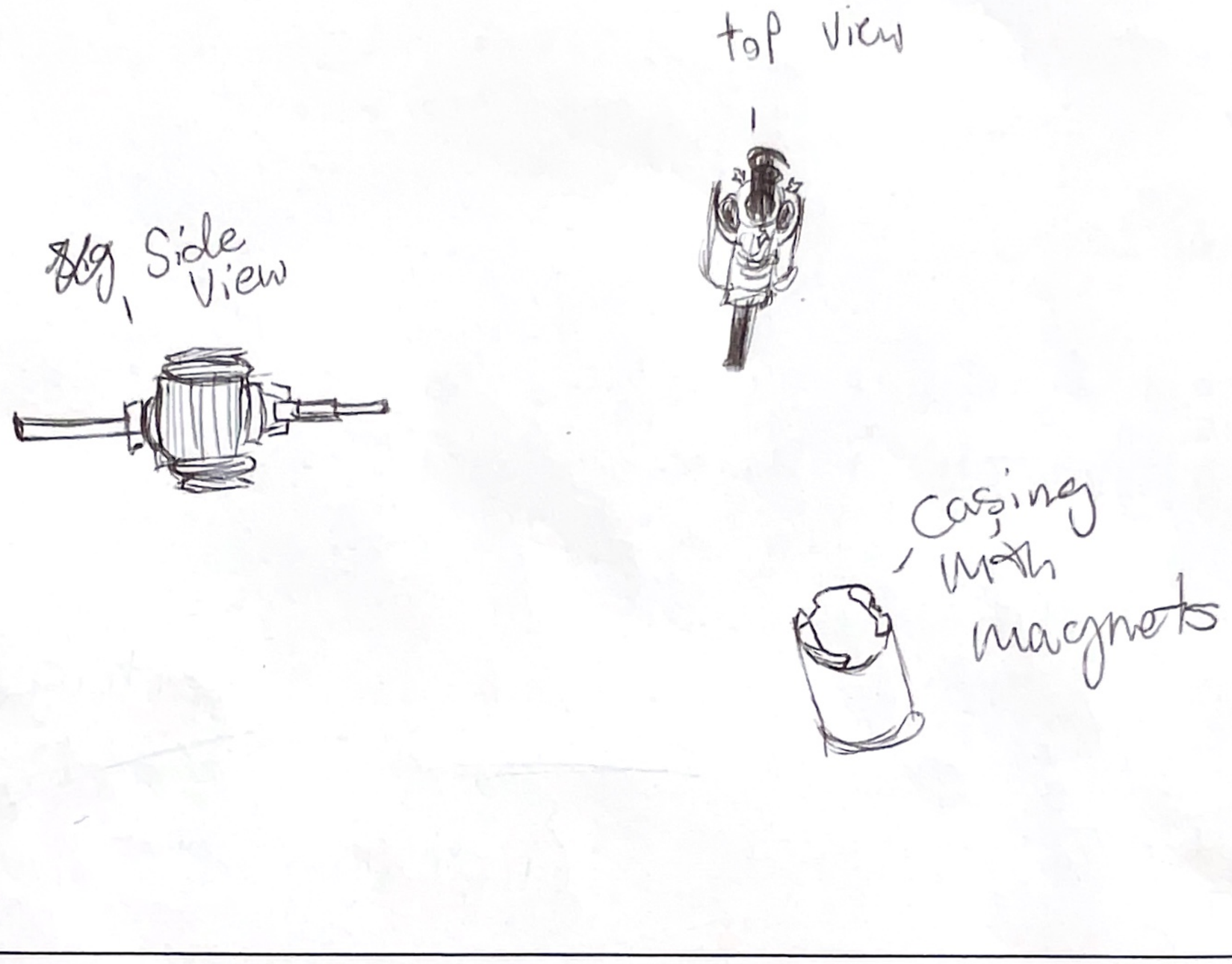
Slow: 0-6 Faster: 10 - 60

3. Talk with your partner. Why do you think that happened?

the motor is creating energy / voltage when spun

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

the ~~Motor~~ wires coiled and magnet and motion

2. What moved at Ngārunui Beach? the magnets and weights

   What turned at Papahua Skatepark? the motor

3. Talk with your partner. Why do you think the meter changed in both tests?

the voltage changed because ~~Kinetic~~ of the magnet made electricity

Lesson 09: How do we generate electricity?

Name: _____ _____

Work with a partner. Visit each place once. If a bench is busy, try another. Do the test, jot down what you see and reset the kit. Step away to talk and write.

Ngārunui Beach

Magnets are attached to the underside of the weights.

1. Keep the weights still. What does the meter read? 0

2. Gently push the weights down a little, then let go. What does the meter do as the weights move? it goes up

3. Talk with your partner. Why do you think that happened?

how the magnet was going made electricity.

Papahua Skatepark

1. Keep the motor still. What does the meter read? 0

2. Spin the shaft of the motor slowly, then faster. What does the meter show?

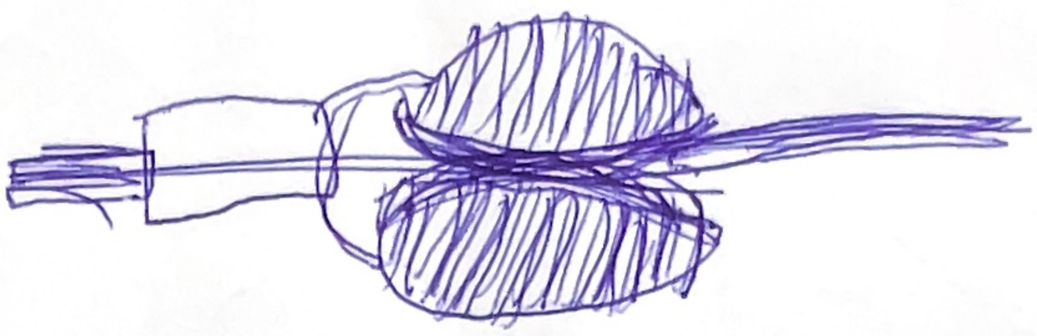
Slow: 0.7 Faster: 706

3. Talk with your partner. Why do you think that happened?

because

Manu Bay

Look inside the motor. Sketch what you see. Draw an arrow to show what turns. Then step away.



After visiting all three places

1. What two parts can you find in both the motor and the Ngārunui Beach setup?

Copper and electricity

2. What moved at Ngārunui Beach? the spring

What turned at Papahua Skatepark? the motor

3. Talk with your partner. Why do you think the meter changed in both tests?

because they both have copper.